

EXPERIENCE

Cisco <i>Software Engineer</i>	San Jose, CA Aug 2024 – Present
• Led a 4-person engineering team to automate the generation of network connectivity maps, reducing the end-to-end process from 2 days to under 4 hours by designing and deploying a serverless pipeline on AWS (Lambda, S3, Sagemaker) • Developed an NLP validation framework that eliminated external LLM API calls, cutting evaluation time by over 95% (from 1.5 hours to under 5 minutes) and boosting validation accuracy from 88% to 98%. • Enhanced data analytics capabilities by optimizing H2O.ai models and developing interactive dashboards with Snowflake SQL	
University of Southern California <i>Researcher</i>	Los Angeles, CA Oct 2024 – April 2025
• Advanced database anomaly detection research by conducting literature review focusing on integrating LLMs with existing diagnostic tools • Improved database anomaly detection capabilities by extending the DBSherlock project through incorporation of GPT-4 • Enhanced database performance diagnostics by developing a methodology combining statistical approaches with GPT-4’s insights	
FICO <i>Data Science Intern</i>	San Diego, CA Apr 2024 – Aug 2024
• Strengthened fraud detection systems by implementing advanced ML models including Isolation Forest, SVM, DBScan, Time Series Analysis, and Autoencoders on data in Hadoop Hive • Expanded anomaly detection capabilities by implementing Classifier-Adjusted Density Estimation (CADE) from scratch based on original machine learning research • Improved model evaluation framework by engineering temporal features across multiple time windows and assessing production viability using Python	
Southwest Airlines <i>Software Engineer Intern</i>	Dallas, TX May 2023 – Aug 2023
• Enhanced internal network automation by developing React.js/Node.js JavaScript web applications for infrastructure team’s use • Contributed to projected annual savings of \$24M by developing website discounting functionality and presenting the solution directly to the CIO • Strengthened software development workflow by participating in Agile methodology including sprint planning, pair programming, and code reviews	
CarsXE <i>Software Engineer Intern</i>	Highland, MD June 2022 – May 2023
• Enhanced web application capabilities by contributing to full-stack development using Vue/Nuxt.js, Next.js, Sentry.io • Accelerated company’s computer vision capabilities by leading other interns license plate recognition implementation using Python, TensorFlow, YOLOv4, OpenCV, and Tesseract OCR via Agile methodology for software development • Improved third-party integration options by developing a Python (pip) package for the company API	

EDUCATION

University of California, San Diego <i>M.S. Machine Learning and Data Science</i>	2024 – 2026
University of California, San Diego <i>B.S. Electrical Engineering</i>	2020 – 2024

PROJECTS

Visual-Inertial SLAM with EKF	SciPy, OpenCV, Estimation Theory, Sensor Fusion
◦ Achieved real-time 6-DoF pose estimation with under 30ms latency by engineering a modular SLAM pipeline fusing IMU and monocular camera data ◦ Attained sub-centimeter positioning accuracy by implementing Extended Kalman Filter for state estimation with sensor fusion ◦ Optimized system performance for real-time applications by designing architecture for feature extraction, pose estimation, and 3D landmark triangulation	
FaceFinder-NamUS	ArcFace, Computer Vision, Deep Learning, Face Recognition
◦ Enhanced missing persons identification capabilities by building a facial recognition and age estimation system for NamUS database ◦ Improved face detection and feature extraction by designing a scalable computer vision pipeline with deep learning models ◦ Increased identification accuracy by 37% over baseline implementations by developing multi-modal matching algorithms	

TECHNICAL SKILLS & ADDITIONAL INFORMATION

- **Tools:** Python, C++, Java, JavaScript, TypeScript, MATLAB, SQL, TensorFlow, PyTorch, React, Flask, OpenCV, Node.js, Elasticsearch, Kafka, OOP, data structures & algorithms, distributed systems, version control (Git), machine learning, CV
- **Languages & Work Eligibility** Fluent in English and Arabic; **Authorized to work in U.S. and U.K (Dual Citizen)**
- **8x Hackathon Winner:** including competitions at **Stanford University, Cambridge University, UCLA, LPL Financial**